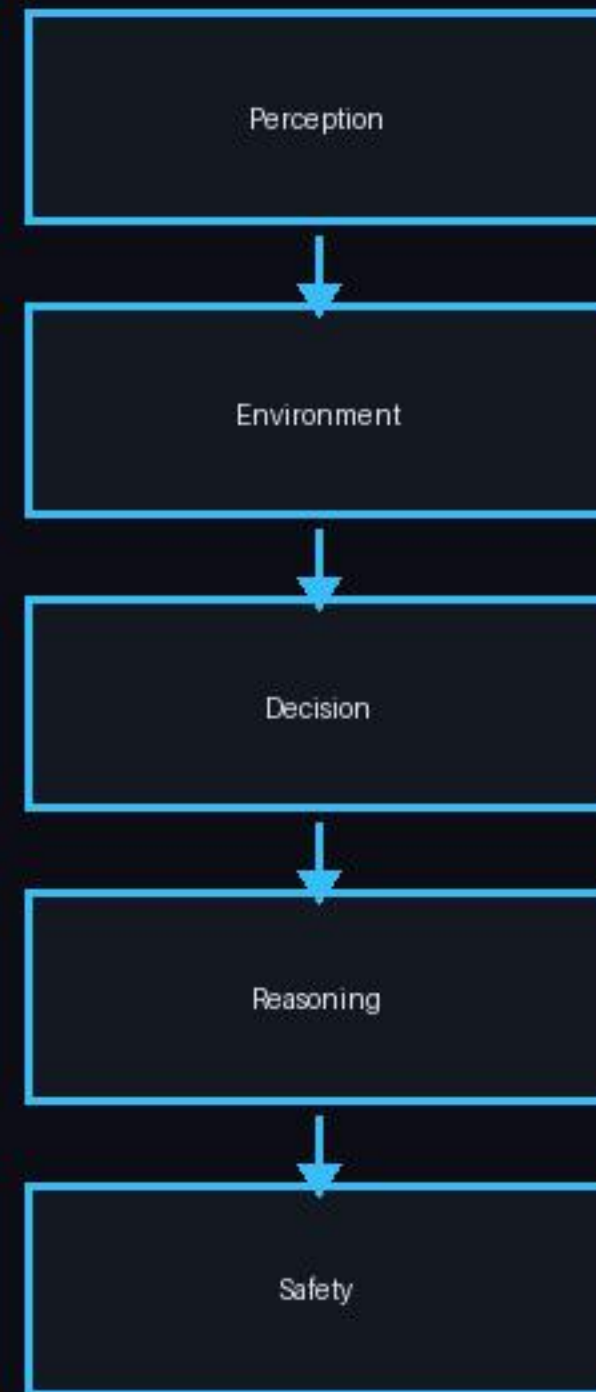


Architecture Overview



Interview Talking Points

▣ Key Contributions

- ▣ - Designed a multi-agent DAG-based orchestration for clear separation of concerns.
- ▣ - Implemented a hybrid perception stack (classical OpenCV lane detection + YOLOv8 object detection).
- ▣ - Built a deterministic safety layer to validate and override LLM-generated suggestions.
- ▣ - Integrated explainability using Gemini while ensuring operational fallbacks.

▣ Metrics to Mention

- ▣ - Detection throughput (fps) on "yolov8n" model and average inference latency.
- ▣ - False positive rates for critical classes (pedestrian, vehicle) on sample dataset.
- ▣ - Safety override incidents per 1,000 frames (demonstrate conservative tuning).

▣ Common Interview Questions & Suggested Answers

Q: Why mix classic CV with deep learning?

- ▣ - A: Classic CV (lane detection) is deterministic, low-latency, and explainable for structured features like lane markings. YOLOv8 provides robust detection for varied objects. Combining both leverages strengths and reduces dependency on only learned features.